



Tuning of LQR-PID controller to control parallel manipulator

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Abstract

The paper presents mathematical modeling and optimal path control of a 3-DOF Maryland manipulator. Three dissimilar and consecutive paths are taken under observation, and control action is performed by linear quadratic regulator (LQR)-based proportional–integral–derivative (PID) controller. To achieve optimal path tracking control, tuning of PID gain parameters is necessary and it is done by optimal selection of weighting matrices of LQR. The evolutionary optimization algorithms like GA and PSO are used in the past for the optimal selection of LQR parameters for tuning of PID gain parameter, but both the methods fail to achieve accurate trajectory tracking control because the simulation results show higher values of fitness function (J), sum square error, integral square error, and integral absolute error between the desired and the actual trajectory for all joint angles, as discussed in the result analysis section. The gray wolf optimizer (GWO) algorithm is then proposed; this method provides optimal values of gain parameters. The simulation results show that the values of all types of error and fitness function for all joint angles are quite lesser than GA and PSO. Hence, the proposed GWO proves better among all and it provides high accuracy in trajectory tracking control with better performance indices. To demonstrate the proposed algorithm, the mathematical simulations are performed as well as the conduction of experimental work.

Keywords Maryland manipulator · Path control · LQR-PID optimization · GA · PSO · GWO · Torques

1 Introduction

With the recent development in the field of robotics, the parallel manipulators have become more important in industries and medical applications, with high stability, precision, and efficiency. The use of controllers is generally utilized in the controller stage in the like manner of pilot training programs [1], fields of clinical [2–5], substantial machine apparatuses [6–8], miniature movement [9], and simplified performing various tasks [10, 11].

Over the last few decades, many researchers came up with new control techniques which provide optimal and

accurate path tracking control, but none of them is comparable with a PID controller due to its simplicity and easy execution. Ren et al. [12] introduced rounded control contingent on the rounded amalgamation technique to improve the imperfection in trajectory tracking.

G. Chaudhary et al. [13] controls the 3-DOF parallel controller using a PID controller; Su et al. [14] proposed nonlinear type of PID controller to compute the control law which increases the coefficient of damping ratio and reduces the path tracking control of a 6-DOF controller. Li and Wang [15] have tuned the PID controller by neural network which enhances the path following control of the manipulator. In spite of numerous of advances in the field of path tracking control, the PID is regularly used as the control method in numerous industries due to its feasibility, straightforwardness, and strength [16, 17].

The four significant shortcomings of PID controllers like computational error, noise degradation, oversimplification, and problems in path tracking control are due to ineffective tuning. This encouraged many researchers to introduce a new optimal control theory to design a linear quadratic regulator (LQR), which reduces the outcomes of state

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equation with minimum control effort [18, 19]. This behavior of LQR prompted many control system designers to implement it for tuning of PID controllers [20].

In the planning of the LQR regulator, the weighting matrices like state (Q) and control (R) should be chosen optimally, in light of the fact that it will influence the control performance of the controller. This encouraged numerous developmental streamlining calculations for the optimal selection of LQR parameters, like gray wolf optimizer (GWO) calculation [21, 22], particle swarm optimizer (PSO) [23, 24], and genetic algorithm (GA) [25, 26]. Lianchao Sheng et al. [27] use GA to advance the additional boundaries of LQR-PID for the direction following control of 3-RRR equal controller. Xuejuan Shao et al. [28] controls the position of the overhead crane by PSO-straight quadratic controller (LQR) PID.

In past, numerous experiments are carried out for the tuning of LQR-PID parameters by evolutionary methods like GA and PSO for path tracking control; the results were satisfactory, but it can be further optimized. The new evolutionary method known as gray wolf optimizer (GWO) [22, 29] is proposed to improve the accuracy in path tracking control by the LQR-PID controller. The results of the proposed method are compared with other past algorithms, which shows that the GWO provides lesser values of overshooting and settling time and minimum path tracking error and produces optimum torque at the active joints and high accuracy in path tracking over PSO and GA. This urged numerous researchers to incline toward improvement in path tracking with GWO by rectifying the problems associated with choosing optimal LQR-PID gain parameters.

In this paper, a 3-DOF manipulator is evolved by Tsai and Stamper [30] in which three progressive trajectories are taken under perception. For precise path tracking [31] of parallel manipulator, control approach such as LQR-PID is developed. The control parameters, such as Q and R of LQR, and the gain parameters of PID are optimized by some popular evolutionary methods like GA, PSO, and proposed GWO. Accordingly, the optimal path and torque of the joints of parallel manipulator are simulated in MATLAB.

The overshoot and settling time, upsides of total square error (SSE), integral square error (ISE), integral absolute error (IAE), and fitness function (J) values are minimum in GWO when contrasted with PSO and GA. Thus, GWO-based LQR-PID regulator shows a lot of steady, effective, precise, and stronger calculations than other ideal tuning calculations.

2 Maryland manipulator

In this paper, Maryland manipulator [30] is taken under perception as presented. An experimental examined manipulator is designed as shown in Fig. 1. It is composed of a moving platform, a fixed base, and three kinematic chains. The moving platform is known as end effector, which is connected by three passive links. Each arm has two connections, and both are associated with one another with revolute joints; the first is an active link and other one is a 4-link parallelogram. Every input link is associated with PMDC motor which is actuated by the rotor of the motor, because of which the moving platform of the examined manipulator moves in three-degree-of-freedom translational motion.

The experimental setup specification with initial values of the examined system is displayed in Table 1.

A layout of the examined framework and the representation of the angular joint are displayed in Fig. 2(a) and (b); the points A_i ($i = 1, 2, 3$) address the culminations of a proper three-sided platform. With the end goal of examination, the reference organize framework (x, y, z) to the fixed platform with its starting point situated at three-sided center O and the z-axis opposite to the stage and the x–y-axes lies on a fixed plane. The beginning of this coordinate central framework is found exactly at the middle P of the moving triangle. One more facilitate framework (x_i, y_i, z_i) is joined to the fixed stage at point A_i to such an extent that the x_i -axis is in accordance with the extended line of OA_i , the y_i -axis.

In the Maryland frameworks, opposite kinematics applied to the reference organize points Cx_i, Cy_i , and Cz_i to figure the upsides of reference joint variable of the

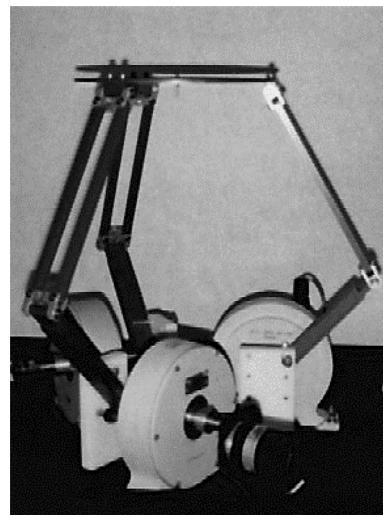
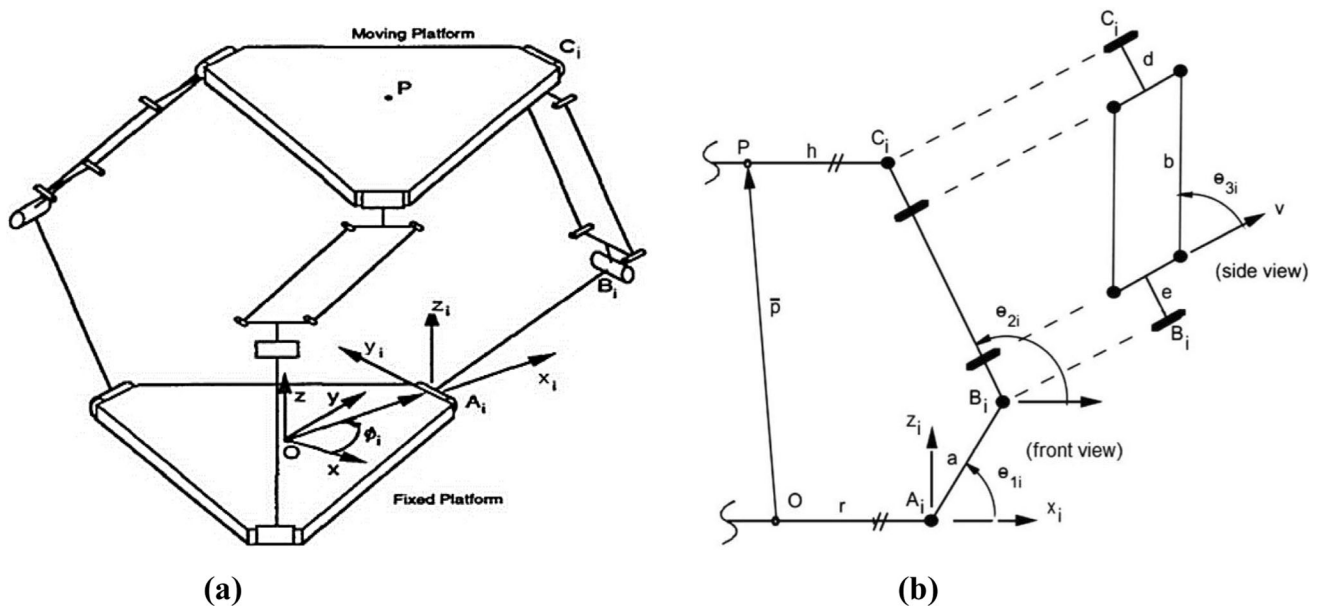


Fig. 1 An experimental setup (image taken in robotics laboratory of NIT, Kurukshetra, by taking permission from Prof. Jyoti Ohri)

Table 1 Specifications of the experimental setup

Symbol	Declaration	Value
ϕ_{1i}	Angle between X - and x_i -axis	[0,120,240] (deg)
a	A_iB_i length	201.8 mm
b	Distance between reference axis and B_iC_i	215.0 mm
e, d	Four neutral bar parallelogram	0.015 mm
r	Base radius	147.0 mm
h	Variable base radius	147.0 mm
m_a	Active link mass	0.196 kg
m_b	Mass of coupled link	0.091 kg
m_p	Movable upper mass	0.424 kg
I_m	Inertia of rotor	0.004 N.m.s ²

**Fig. 2** **a** Configuration of Maryland manipulator [31]. **b** Angular joints [31]

portable base. Articulation (1) and (2) of Cartesian coordinates (x_i, y_i, z_i) of desired trajectories are transformed into joint angles by inverse kinematic.

$$\begin{bmatrix} a \cos(\theta_{1i}) + b \sin(\theta_{3i}) \cos(\theta_{1i} + \theta_{2i}) \\ b \cos \theta_{3i} \\ a \sin(\theta_{1i}) + b \sin(\theta_{3i}) \sin(\theta_{1i} + \theta_{2i}) \end{bmatrix} = \begin{bmatrix} C_{xi} \\ C_{yi} \\ C_{zi} \end{bmatrix} \quad (1)$$

where

$$\begin{bmatrix} C_{xi} \\ C_{yi} \\ C_{zi} \end{bmatrix} = \begin{bmatrix} \cos(\phi_i) & \sin(\phi_i) & 0 \\ -\sin(\phi_i) & \cos(\phi_i) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} \begin{bmatrix} h - r \\ 0 \\ 0 \end{bmatrix} \quad (2)$$

Two solutions of joint angle θ_{3i} are obtained by solving the second row from the matrix expression (1):

$$\theta_{3i} = \pm \cos^{-1} \left(\frac{C_{yi}}{b} \right) \quad (3)$$

θ_{3i} is dictated by utilizing articulation (3); the settled articulation of θ_{2i} is calculated by adding $(C_{xi})^2$, $(C_{yi})^2$, and $(C_{zi})^2$.

$$\theta_{2i} = \pm \cos^{-1}(k) \quad (4)$$

where $k = \frac{C_{xi}^2 + C_{yi}^2 + C_{zi}^2 - a^2 - b^2}{2ab \sin(\theta_{3i})}$. As θ_{3i} and θ_{2i} are acquired which are displayed in (3) and (4), arrangement of θ_{1i} is gotten by utilizing (1), (3), and (4) as composed beneath:

$$\theta_{1i} = \pm \cos^{-1} \left(\frac{C_{zi}k_2 + C_{xi}k_1}{k_1^2 + k_2^2} \right) \quad (5)$$

where $k_1 = a + b \sin(\theta_{3i}) \cos(\theta_{2i})$ and $k_2 = b \sin(\theta_{3i}) \sin(\theta_{2i})$.

Because of the convoluted kinematics of the manipulator, the impelling torques are portrayed and acquired by using the Lagrangian method. The generalized torque equation of motion of moving platform is defined in (6):

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} + \sum_{k=1}^3 \lambda_k \frac{\partial \Gamma_k}{\partial \dot{q}_i} = Q_i \text{ for } i = 1 \text{ to } 6 \quad (6)$$

$$\text{where } q_i = \begin{cases} P_x, P_y, P_z, i = 1, 2, 3 \\ \theta_{11}, \theta_{12}, \theta_{13}, i = 4, 5, 6 \end{cases} \quad Q_i = \begin{cases} F_i, i = 1, 2, 3 \\ T_{i-3}, i = 4, 5, 6 \end{cases}$$

To such an extent that F_i is the outer power in x , y and z segments applied on the moving stage, T_{i-3} shows the i th connect frictional forces in the θ_{1i} heading, Q_i indicates the summed up outside power, λ_k addresses the Lagrange multiplier, and Γ_k is signified by requirement condition.

By rewriting the above torque equation, the three torque equation at three dynamic joints ($\theta_{11}, \theta_{21}, \theta_{31}$) of the Maryland manipulator is obtained by applying Lagrangian method of movement, as displayed in (7, 8, and 9):

$$T_1 = \left(I_m + \frac{1}{3} m_a a^2 + m_b a^2 \right) \ddot{\theta}_{11} \left(\frac{1}{2} m_a + m_b \right) g_c a \cos(\theta_{11}) - 2a\lambda_1 * [(p_x \cos(\theta_1) + p_y \sin(\theta_1) + h - r) * \sin(\theta_{11}) - p_z \cos(\theta_{11})] \quad (7)$$

$$T_2 = \left(I_m + \frac{1}{3} m_a a^2 + m_b a^2 \right) \ddot{\theta}_{12} \left(\frac{1}{2} m_a + m_b \right) g_c a \cos(\theta_{12}) - 2a\lambda_2 * [(p_x \cos(\theta_2) + p_y \sin(\theta_2) + h - r) * \sin(\theta_{12}) - p_z \cos(\theta_{12})] \quad (8)$$

$$T_3 = \left(I_m + \frac{1}{3} m_a a^2 + m_b a^2 \right) \ddot{\theta}_{13} \left(\frac{1}{2} m_a + m_b \right) g_c a \cos(\theta_{13}) - 2a\lambda_3 * [(p_x \cos(\theta_3) + p_y \sin(\theta_3) + h - r) * \sin(\theta_{13}) - p_z \cos(\theta_{13})] \quad (9)$$

The above torques are generated at the three joints of the manipulator by three PMDC. The motor equation in time domain, which drives the moving platform, is shown in Eq. (10):

$$G_{DC}(s) = \frac{\theta(s)}{V(s)} = \frac{K_t}{s[(Ls + R)(J_ms + B_m) + K_v K_t]} \quad (10)$$

where B_m means the damping coefficient, J_m indicates the inaction of the motor and moving arm, K_v indicates the electromotive power consistent of the motor, K_t addresses the force persistent of the rotor, L_s demonstrates the inductance of the winding, R characterizes the obstruction of the motor winding, and V_a indicates the excited voltage.

3 Linear quadratic regulator (LQR) formulation

The linear quadratic regulator is a magnificent technique that utilizes state-space input for streamlining direct frameworks; because of this property, it is widely used in the industry. The point when the Maryland framework is invigorated with an excitation $r(t)$, having a control law $u(t)$ and yield $y(t)$, is displayed in Fig. 3. The transfer function in state-space representation of the manipulator is written as follows:

$$\frac{Y(s)}{U(s)} = \frac{K}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{2.2929}{s^2 + 0.0068s + 1.6231} = -\frac{E(s)}{U(s)} \quad (11)$$

where $W(t)$ signifies noise in the unique framework model. At the point when the framework input is made zero such that $r(t) = 0$, the path following issue of the framework comes to regulator. Assuming there are no changes in the reference input, the modified equation of (11) is rewritten in (12):

$$[S^2 + 2\zeta\omega_n s + \omega_n^2] E(s) = -kU(s) \quad (12)$$

Equation (12) is transformed in time domain by taking inverse Laplace transform, as written as follows:

$$\ddot{e} + 2\zeta\omega_n \dot{e} + \omega_n^2 e = -ku \quad (13)$$

$$\dot{x}_3 + 2\zeta\omega_n x_3 + \omega_n^2 x_2 = -ku \quad (14)$$

By taking the help of the above equations, the state-space model of Maryland manipulator is formulated in (15):

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -K \end{bmatrix} u \quad (15)$$

From the above equation, the system matrix and input matrix of manipulator are obtained as shown in (16):

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1.6231 & -0.0068 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ -2.2929 \end{bmatrix} \quad (16)$$

The main purpose to find the solution of LQR problem is to reduce the value of fitness (quadratic cost) function, as written in (17):

$$J(u) = \int_0^\infty [x^T(t) Q x(t) + u^T(t) R u(t)] dt \quad (17)$$

To achieve minimum value of the above function, the state feedback control law is applied, as shown as follows:

$$u(t) = -R^{-1} B^T P x(t) = -F x(t) \quad (18)$$

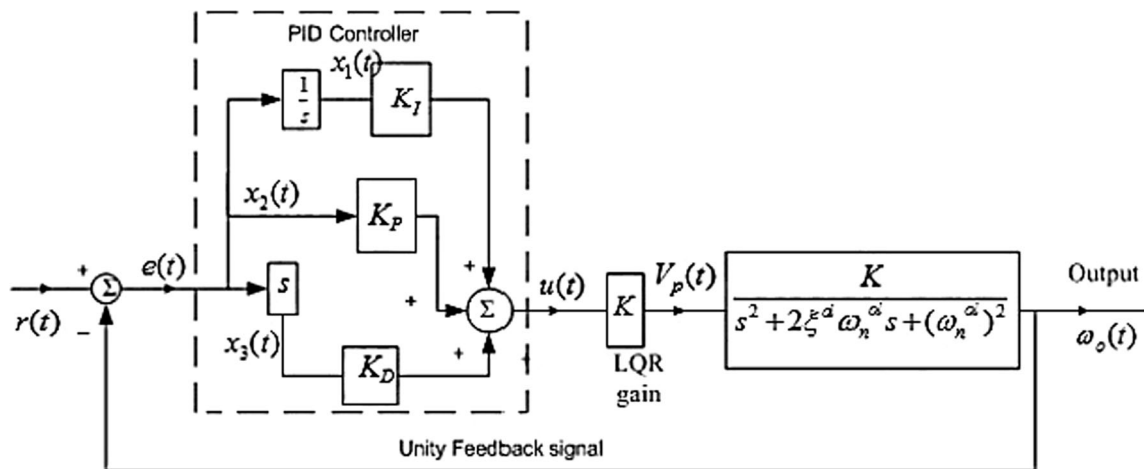


Fig. 3 LQR formulation of PID controller for the second-order manipulator

The solution of Eqs. (17) and (18) yields the values of Q and R matrices. The matrix Q is said to be semi-definite positive, if it satisfies the given condition such that $Q_1, Q_2, Q_3 \geq 0$ and R matrix should be positive integer. These matrices must be optimal in nature in order to achieve different optimization algorithms applied in the past. The Q and R matrices are written as follows:

$$Q = \text{diag} [Q_1, Q_2, Q_3] R = r \quad (19)$$

For the optimal selection of Q and R matrices, there must be lower and upper bound.

$$Q = \text{diag} [0, 0, 0] \text{ to } Q = \text{diag} [500, 500, 500] \text{ and } R = 0.01 \quad (20)$$

After obtaining optimal values of Q and R matrices and substituting them in continuous algebraic Riccati Eq. (21), the symmetric positive matrix P is obtained, as shown in (22):

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (21)$$

where

$$P = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{12} & P_{22} & P_{23} \\ P_{13} & P_{23} & P_{33} \end{bmatrix} \quad (22)$$

Then after, the state feedback (F) is calculated and formulated as follows:

$$F = -R^{-1} B^T P = -R^{-1} \begin{bmatrix} 0 & 0 & K \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{12} & P_{22} & P_{23} \\ P_{13} & P_{23} & P_{33} \end{bmatrix} \quad (23)$$

$$F = -R^{-1} K \begin{bmatrix} P_{13} & P_{23} & P_{33} \end{bmatrix} = \begin{bmatrix} K_i & K_p & K_d \end{bmatrix} \quad (24)$$

From the above equation, the PID gain parameters are rewritten in (25):

$$K_i = -R^{-1} K P_{13}, K_p = -R^{-1} K P_{23}, K_d = -R^{-1} K P_{33} \quad (25)$$

The lower and upper bound of the optimal PID gain parameters from the simulation results are shown as follows:

$$K_i = [0 : 80], K_p = [0 : 100] \text{ and } K_d = [0 : 50]. \quad (26)$$

4 Description of evolutionary optimization algorithms

For accurate path tracking [32] of parallel manipulator, LQR-PID control approach is formulated. The optimal tuning of gain parameters of PID is decided by the optimal selection of Q and R weighting matrices of LQR. Once the optimal value of K_p, K_i , and K_d is achieved, the controller provides actual (optimal) path by generating optimal torque at the joints.

The values of fitness function and errors such as SSE, ISE, and IAE between the desired and actual path are then calculated. The values of fitness function and errors decide the ability of optimization methods that provide accurate trajectory tracking control.

The evolutionary optimization algorithms are introduced for the optimal selection of LQR parameters for tuning of PID gain parameters because they will affect the control performance. The outcomes of different algorithms are briefly described in this section.

4.1 Genetic algorithm (GA)

It is the most famous and most demanding method. Numerous examinations have been done on tuning of LQR-PID by GA [33, 34]. It is a heuristic search technique that consistently begins with no thought of the right

arrangement of the issue. The genetic algorithm measures consistently with an arbitrary populace that is shaped by strings of pairs called chromosomes. Its existence is constantly checked, and furthermore its endurance for its existence is tested each time by checking the fitness function, till the best combination of chromosomes, as displayed in Fig. 4.

This method is applied for trajectory tracking control of the Maryland manipulator. The optimal selection of LQR-PID gain parameters is done, but the values of fitness function and error between the desired and the actual trajectory for all joint angles are quite high, as discussed in the result analysis section. Hence, GA fails to achieve accurate trajectory tracking control.

4.2 Particle swarm optimizer (PSO)

PSO [35] is mainstream among any remaining meta-heuristic calculations for optimization, which is influenced by the social demeanor of the birds. In the present situation, the PSO optimization calculation has been used in numerous applications like numerical confusions, logical extension issues, and numerous other unique designing fields [36, 37].

The PSO calculation begins by creating an underlying occupant (swarm) including haphazardly delivered particles inside an instatement region. Subjective area and speed dispensed particles search the best arrangement by exploring in the issue search region. Particles at first beginnings look through the situation in the issue search zone by means of characterized details. The PSO calculations are clarified momentarily in the stream diagram as

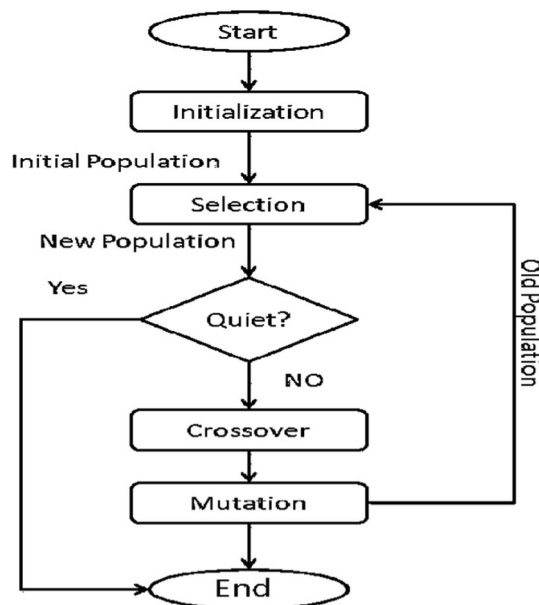


Fig. 4 Flowchart for GA procedure

displayed in Fig. 5. In search space, each particle adjusts its position in three dimensions; after every iteration, the particles upgrade their current position and velocity and move to their best positions; this process continues till the best solution is reached by reducing the fitness value (J).

$$J = \int_0^{\infty} [Q_1 x_1^2(t) + Q_2 x_2^2(t) + Q_3 x_3^2(t) + R u^2(t)] dt \quad (27)$$

In each iteration, the i th particle adjusts its position and velocity, as written as follows:

$$v = [v_{i1}, v_{i2}, v_{i3}, v_{in}] \& y = [y_{i1}, y_{i2}, y_{i3}, y_{in}] \quad (28)$$

where $i = 1, 2, 3, \dots, n$

$$v_i^{t+1} = wv_i^t + c_1 r_1 (P_{i,best}^t - y_i^t) + c_2 r_2 (G_i^t - y_i^t) \quad (29)$$

$$y_i^{t+1} = y_i^t + v_i^{t+1} \quad (30)$$

v_i^{t+1} denotes the current velocity, v_i^t denotes the previous velocity, y_i^{t+1} shows the current position, y_i^t shows the previous position, $P_{i,best}^t$ denotes the particle best position, and G_i^t denotes the global best.

This algorithm is implemented for trajectory tracking control of Maryland manipulator. The optimal selection of LQR-PID gain parameters is done, but in the simulation results, the values of fitness function and error for all joint angles are slightly high, as discussed in the result analysis section. Hence, PSO fails to achieve accurate trajectory tracking control.

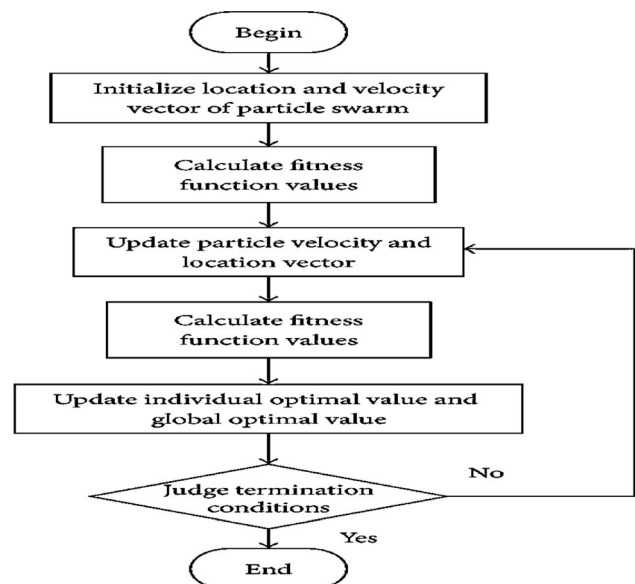


Fig. 5 PSO process

4.3 Proposed gray wolf optimizer (GWO)

GWO was presented by Mirjalili et al. [38] in 2014, which relies upon the looking and following method of gray wolves based on their progressive system of initiative in their gathering; numerous investigations [39–43] have been done in its writing wherein the GWO calculation is inspected exhaustively. Optimal Q and R matrices are produced by reducing the objective/fitness function. The proposed methodology is shown in the form of flowchart in Fig. 6.

$$J = \int_0^{\infty} [Q_1 x^2(t) + Q_2 \dot{x}^2(t) + Q_3 \ddot{x}^2(t) + R u^2(t)] dt \quad (31)$$

There are four particular social events according to their solidarity between each other in the chain of significance, which are according to the accompanying: alpha (α), beta (β), delta (δ), and omega (ω) independently.

The alpha (α) wolf in the chain of significance is the pioneer among any remaining gatherings who make decisions about looking, pursuing and different starts like resting time and locales in the wild, etc. Beta (β) wolf is put at the second situation in the chain of significance as a partner of alpha (α) wolf who is furthermore endorsed to make decisions. Like alpha (α) and beta (β) wolves, delta (δ) is in the most insignificant situation, however, prevailing more than omega (ω).

In GWO, encompassing a prey by the assistance of α , β , and δ is the significant motivation. It is expected to be that α , β , and δ are having genuine data about the current position and area of the prey from the whole arrangements. Along these lines, the essential fittest arrangements are

adjusted until newer best arrangements are produced, which can be systematically dug in as underneath. From that point onward, the surrounding conduct of the gray wolves is reproduced, as displayed in (32) and (33):

$$\vec{W}(t+1) = \vec{W}_r(t) - \vec{M} \cdot \vec{Q} \quad (32)$$

From Eq. (32), $\vec{Q}$ can be formulated as shown as follows:

$$\vec{Q} = \left| \vec{S} \cdot \vec{W}_r(t) - \vec{W}(t) \right| \quad (33)$$

Here, $\vec{W}$ means the gray wolf present position, $\vec{W}_r$ signifies the prey's real position, $\vec{Q}$ indicates the gray wolf's real circle of prey, $\vec{M}$ and $\vec{S}$ considered as the coefficient vectors usually, and ' t ' means the quantity of iterations for the fittest arrangement. The coefficient vectors $\vec{M}$ and $\vec{S}$ can be achieved by the condition (34):

$$\vec{M} = 2\vec{m} \cdot \vec{x}_1 - \vec{m}, \quad \vec{S} = 2\vec{x}_2 \quad (34)$$

where ' $\vec{m}$ ' will be unquestionably decreased from 2 to 0 and $\vec{x}_1$ and $\vec{x}_2$ are the self-assertive vectors from [0, 1]. The variable ' $\vec{m}$ ' is changed in each simulation esteem inside the reach from 0 to 2 according to (35):

$$\vec{m} = 2 - t \left[\frac{2}{t_{\max}} \right] \quad (35)$$

$t_{\max}$ addresses the complete number of emphases allowed. It is assumed that large amount of data are needed to find the present location of prey through a definitive arrangement of α , β , and δ . The wolves' new position is refreshed, in view of the best arrangements that can be acquired as displayed in (36):

$$\begin{aligned} \vec{W}_1 &= \left| \vec{W}_\alpha(t) - \vec{M}_1 \cdot \vec{Q}_\alpha \right|, \vec{W}_2 \\ &= \left| \vec{W}_\beta(t) - \vec{M}_2 \cdot \vec{Q}_\beta \right| \& \vec{W}_3 = \left| \vec{W}_\delta(t) - \vec{M}_3 \cdot \vec{Q}_\delta \right| \end{aligned} \quad (36)$$

where $\vec{Q}_\alpha$, $\vec{Q}_\beta$, and $\vec{Q}_\delta$ are obtained from the above equations, as follows:

$$\begin{aligned} \vec{Q}_\alpha &= \left| \vec{S}_1 \cdot \vec{W}_\alpha - \vec{W} \right|, \\ \vec{Q}_\beta &= \left| \vec{S}_2 \cdot \vec{W}_\beta - \vec{W} \right| \& \vec{Q}_\delta = \left| \vec{S}_3 \cdot \vec{W}_\delta - \vec{W} \right| \end{aligned} \quad (37)$$

In view of all the above conditions, the new solution for next emphasis can be formulated as

$$\vec{W}(t+1) = \frac{\vec{W}_1 + \vec{W}_2 + \vec{W}_3}{3} \quad (38)$$

The new position of wolves refreshes ceaselessly until the most extreme emphasis is accomplished. On the off chance that regardless the most iteration or ideal

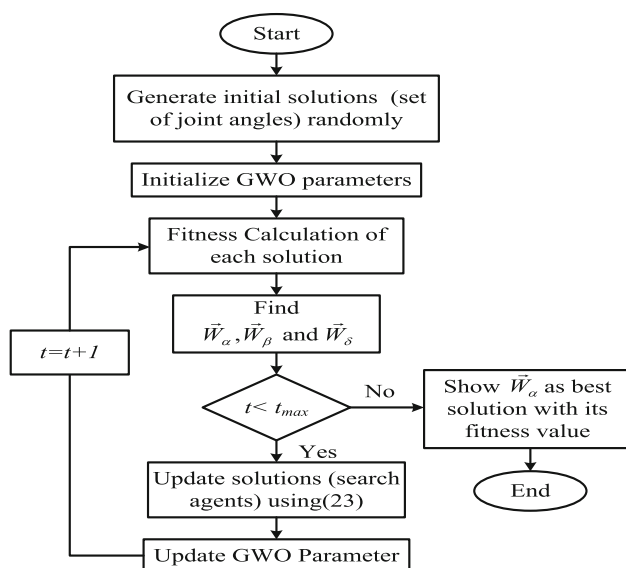


Fig. 6 Optimization process of GWO

arrangement does not accomplished, then, at that point no new solution will be found and for which fitness assessment happens, and afterward depending on the fittest among past arrangement, further changes will be done on ordinary premise.

Finally,

$$Q = \text{diag} [Q_1, Q_2, Q_3] = \text{diag} [\vec{W}_1, \vec{W}_2, \vec{W}_3] \quad (39)$$

This algorithm is implemented for trajectory tracking control of the Maryland manipulator. The optimal selection of LQR-PID gain parameters is obtained and analyzed in the simulation; the results show that the values of fitness function and error on all joint angles are lesser than GA and PSO, as discussed in the result analysis section. Hence, the proposed GWO proves better among all and it provides high accuracy in trajectory tracking control.

5 Specification and mathematical modeling of LQR-PID controller

In this segment, the three evolutionary optimization algorithms are implemented for the optimal selection of LQR parameters for tuning of PID gain parameters because it will affect the control performance.

The errors like SSE, ISE, and IAE between the desired and the actual path depend upon the optimal gain parameters of LQR-PID, and the fitness function (J) depends upon the types of error.

For acquiring minimum values of fitness function, three evolutionary algorithms are applied. The fitness function (J) is formulated as follows:

$$J(k_i, k_p, k_d) = w_1 \text{SSE} + w_2 \text{ISE} + w_3 \text{IAE} \quad (40)$$

$$\text{SSE} = \frac{1}{N} \sum_{i=1}^N (e_i)^2, \text{ISE} = \int_0^{\infty} [e(t)]^2 dt, \text{IAE} = \int_0^{\infty} [e(t)] dt$$

where $w_1 = 1$, $w_2 = 2$, and $w_3 = 1$ are the picked weighting factor [44], as selected by the designer. The boundaries of the picked calculations are given in Table 2.

The evolutionary algorithms are coded and simulated in MATLAB for optimal selection of LQR-PID gain parameters. The optimization of LQR-PID acquires grids, as arranged in Table 3.

6 Simulation and experimental results analysis

The recommended algorithms for path following control of the Maryland framework are executed in MATLAB. To test the recommended control algorithms, we consider

Table 2 Evolutionary optimization algorithms' specifications

GWO	Domain of search for $\vec{M}$	[−1000,1000]
	Search boundaries for $\vec{m}$	[−1,1]
	Search boundaries for $\vec{S}$	[0,1000]
	Search length	2
	Overall search particles	20
PSO	Total iteration	100
	Total particles participated	50
	Velocity-associated variables	3
	Inaction load W_{initial}	0.2
	W_{final}	0.9
GA	Area finest value (c_1)	2
	Personal finest value (c_2)	2
	Total iteration	100
	Participated elements	4
	Most noteworthy production	160
	Occupants dimension	30
	Censorious	0.7
	Alteration	0.3
	Total iteration	100

Table 3 Optimized gain parameters

		GWO	PSO	GA	Un-optimized
LQR	Q_1	369.32	150.06	45.87	1.0
	Q_2	281.19	170.27	70.56	1.0
	Q_3	147.63	200.83	60.63	1.0
	R	0.08	1	1.5	1.0
PID	K_i	67.94	12.24	5.52	1.0
	K_p	96.11	22.14	10.24	1.22
	K_d	43.00	14.20	6.38	1.02

three separate back-to-back positions in x-, y-, and z-axes as displayed in 41, 42, and 43:

$$\left. \begin{array}{l} P_x = 0 \\ P_y = 0 \\ P_z = 0.40 + t^2/1200 \end{array} \right\} \text{ if } t \leq 5 \quad (41)$$

$$\left. \begin{array}{l} P_x = 0.09 \cos(t) \\ P_y = 0.09 \sin(t) \\ P_z = 0.40 \end{array} \right\} \text{ if } 5 < t \leq 15 \quad (42)$$

$$\left. \begin{array}{l} P_x = 0 \\ P_y = 0 \\ P_z = 0.40 - t^2/1200 \end{array} \right\} \text{ if } t > 15 \quad (43)$$

Utilizing the above position equation in x , y , and z directions, the pictorial portrayal of manipulator positional directions is displayed in Fig. 7.

The 3-DOF Maryland manipulator has three joint angles ($\theta_{11}, \theta_{12}, \theta_{13}$). LQR-PID controller controls the joint angles of the manipulator, so that the actual path followed by the manipulator must be equal to the desired path. Three evolutionary algorithms (GA, PSO, and GWO) are applied for optimal tuning of LQR-PID controller parameters.

6.1 Comparison of path tracking control of all joints by GA, PSO, and proposed GWO

The actual and desired path tracking of active joint angle (θ_{11}) along with the tracking error between them is graphically plotted in Fig. 8.

As displayed in the above figure, un-optimized LQR-PID regulator controls the path of the active joint angle of

the Maryland manipulator but fails to follow the desired path due to the external disturbance that influenced the system.

If at any situation, the motor does not produce the desired torque at the active joints, the path tracking error between the desired and actual path will be maximum. So, to minimize the path tracking error and to improve the path tracking performance, three unique evolutionary algorithms, i.e., Ga, PSO, and proposed GWO, are applied.

The values of fitness function (J), SSE, ISE, and IAE between the desired and actual path for Ga, PSO, and proposed GWO and the performance indices of joint angle (θ_{11}) are compared and displayed in Table 4.

GA, PSO, and proposed GWO algorithm produce optimal actual path, which is compared with the desired path with the tracking error for the first passive angle (θ_{12}). The simulation results are displayed in Fig. 9.

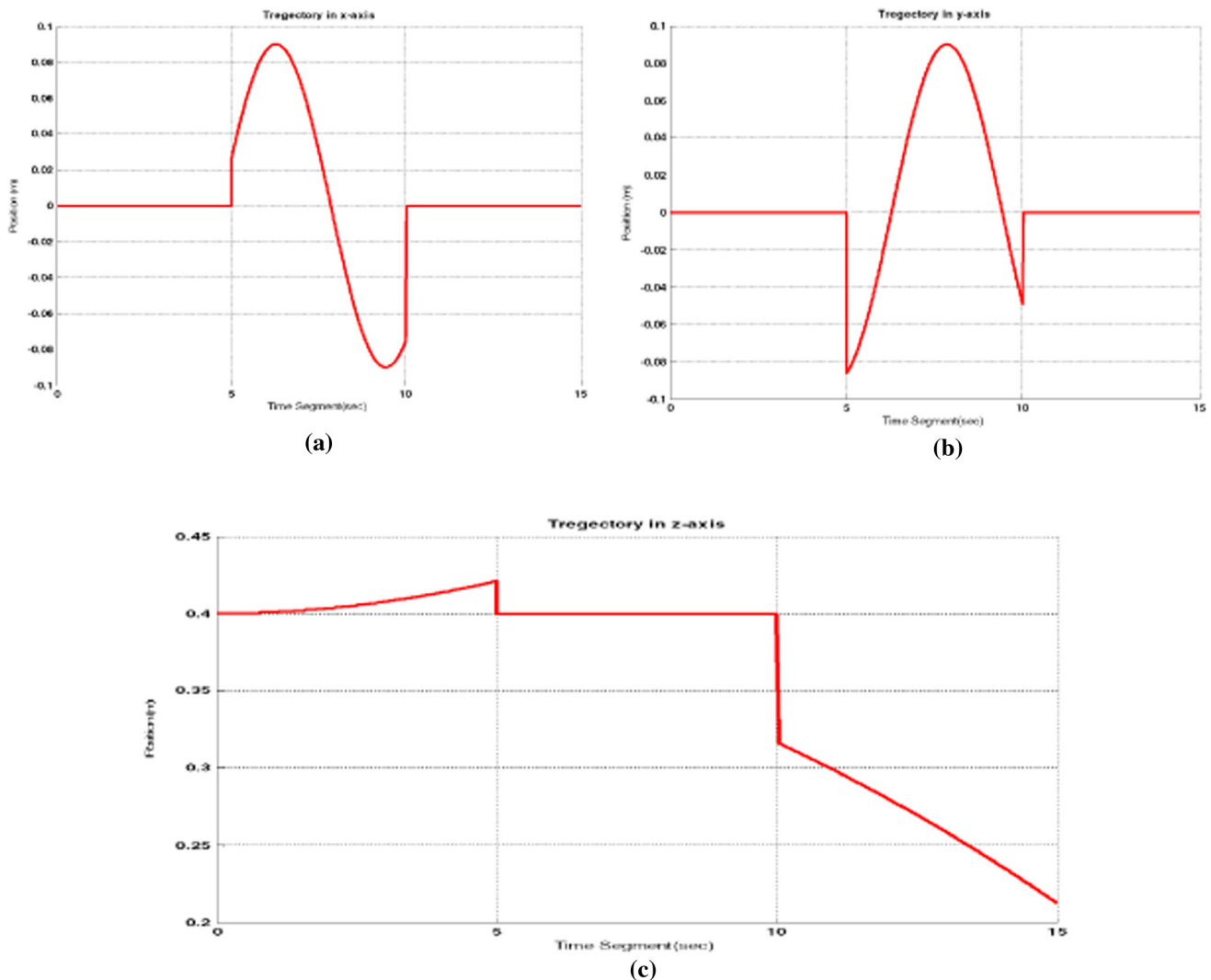


Fig. 7 Three sequential trajectories in **a** x, **b** y, and **c** z directions

Fig. 8 Current and desired paths of active joint (θ_{11}) with tracking error between them for GA, PSO, GWO algorithms and experimental, respectively

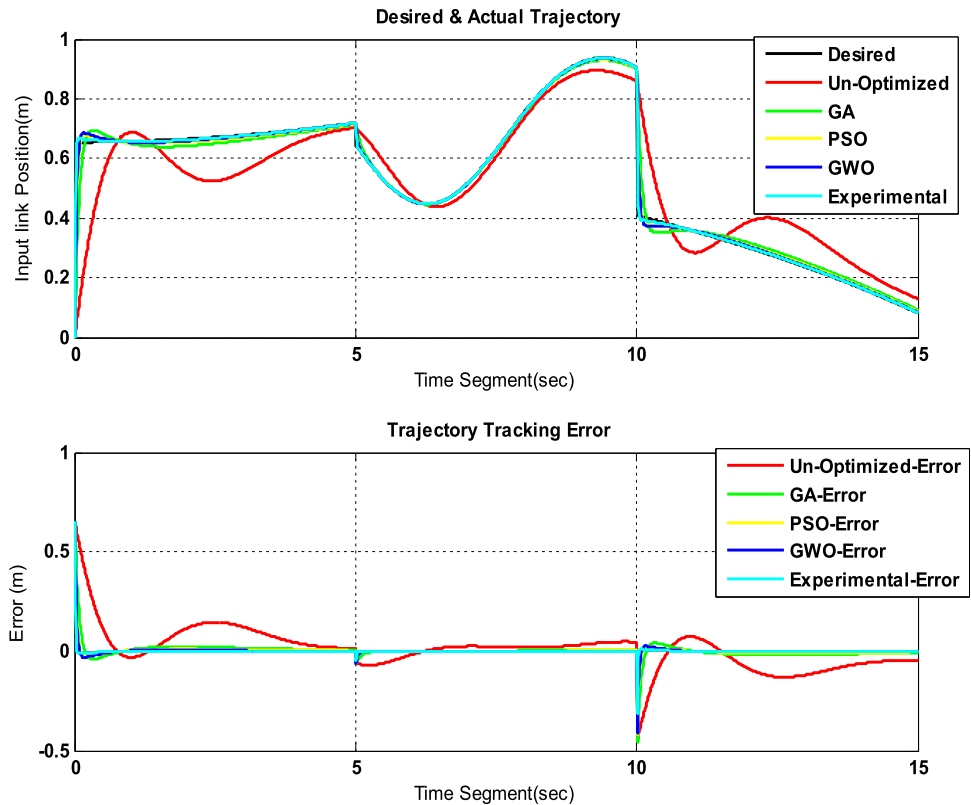


Table 4 Comparison of error and fitness function of joint angle (θ_{11})

Trajectory Tracking Performance Indices Comparison				
Error	■ Un-Optimize ■ GA ■ PSO ■ GWO ■ Experimental			
	SSE	ISE	IAE	Jmin
■ Un-Optimize	16.85	0.87	1.63	18.08
■ GA	1.395	0.096	0.172	3.429
■ PSO	0.895	0.0531	0.0982	1.524
■ GWO	0.512	0.006	0.0213	0.0785
■ Experimental	0.526	0.0062	0.0226	0.0802

Joint Angle (Θ_{11})

To minimize the path tracking error and to balance out the path tracking performance, three unique evolutionary algorithms are applied. The values of fitness function (J), SSE, ISE, and IAE among the current and desired paths for Ga, PSO, and proposed GWO and the performance indices of joint angle (θ_{12}) are compared and displayed in Table 5.

Evolutionary algorithms are applied, which produce optimal actual path, which is compared with the desired

path with the tracking error for second passive angle (θ_{13}). The simulation results are graphically displayed in Fig. 10.

The values of fitness function (J), SSE, ISE, and IAE among the actual and desired paths for Ga, PSO, and proposed GWO and the performance indices of trajectory tracking control of joint angle (θ_{13}) are compared and displayed in Table 6.

Fig. 9 Current and desired paths of joint (θ_{12}) with tracking error between them for GA, PSO, GWO algorithms and experimental, respectively

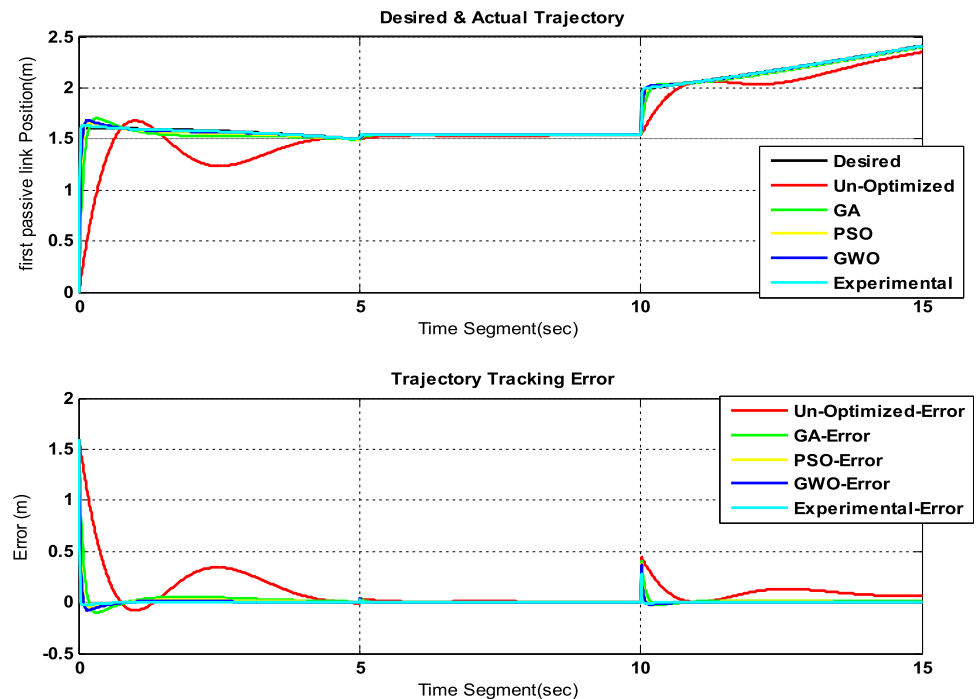


Table 5 Comparison of error and fitness function of joint angle (θ_{12})

Trajectory Tracking Performance Indices Comparison				
Error	■ Un-Optimized ■ GA ■ PSO ■ GWO ■ Experimental			
	SSE	ISE	IAE	Jmin
■ Un-Optimized	42.23	5.41	3.48	44.13
■ GA	3.435	2.071	1.893	9.658
■ PSO	1.809	0.921	0.761	6.669
■ GWO	1.132	0.011	0.014	1.158
■ Experimental	1.154	0.013	0.016	1.161

Joint Angle (θ_{12})

The path tracking examination of the recommended methods is compared with various error esteems, as displayed in Figs. 8, 9, and 10. The proposed GWO algorithm tracks the desired path with very less error and also minimum value of fitness function as compared with PSO and GA. Hence, it is clear that GWO provides accurate path tracking control.

6.2 Comparison of optimal torque control of active joint by GA, PSO, and GWO

For ideal path tracking, ideal torque is required which is produced by the PMDC motor, which is connected at the active joints of the manipulator. The LQR-based PID regulator is tuned to acquire desired torque from the motor by applying GA, PSO, and proposed GWO algorithms. Both the acquired and desire torques alongside the error

Fig. 10 Current and desired paths of joint (θ_{13}) with tracking error between them for GA, PSO, GWO algorithms and experimental, respectively

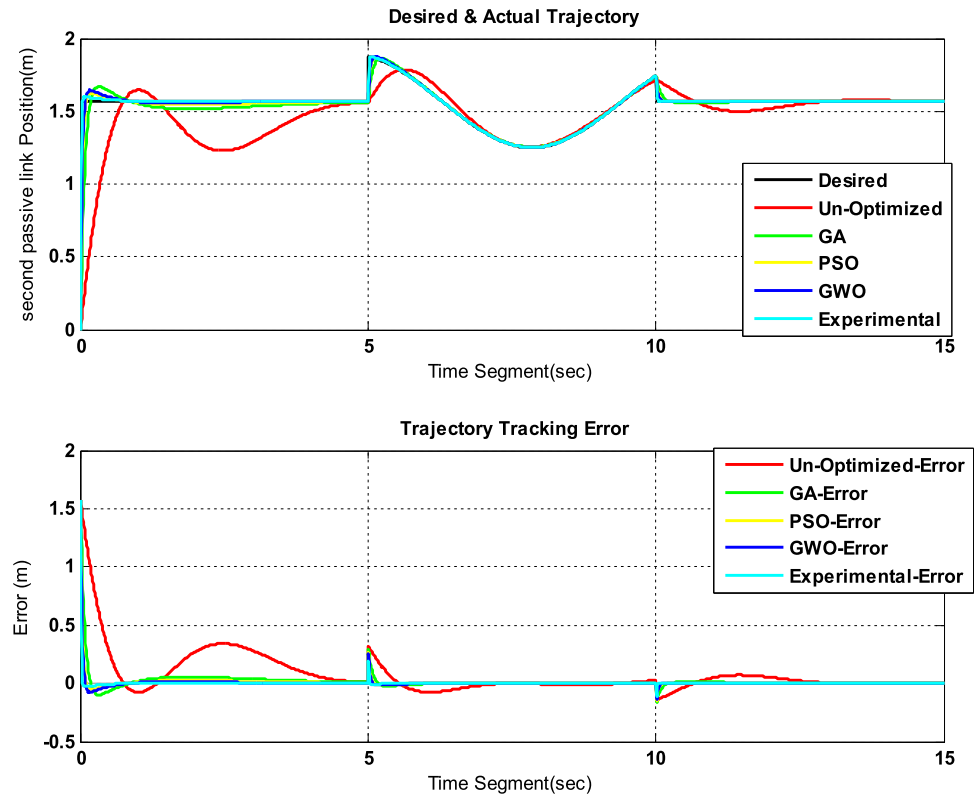


Table 6 Comparison of error and fitness function of joint angle (θ_{13})

Trajectory Tracking Performance Indices Comparison				
Error	■ Un-Optimized ■ GA ■ PSO ■ GWO ■ Experimental			
	SSE	ISE	IAE	Jmin
■ Un-Optimized	63.86	21.62	35.48	65.97
■ GA	5.97	12.29	16.89	19.25
■ PSO	2.78	4.94	7.56	5.43
■ GWO	2.47	0.02	0.01	2.51
■ Experimental	2.515	0.031	0.013	2.634

Joint Angle (θ_{13})

between the ideal and real torque for active link are graphically plotted in Fig. 11.

In view of minimum fitness function ($J_{\min}$), ISE, IAE, and SSE values as for the torque delivered by the PMMDC to the active link (θ_{11}), the proposed GWO-tuned LQR-PID regulator creates ideal torque to accomplish desired path.

The performance indices of torque at joint angle (θ_{11}) are compared and displayed in Table 7.

The graphical plots provided in Figs. 8, 9, 10, and 11 show that the external disturbances and uncertainties present in the manipulator that affects the trajectory tracking control and the actual generated trajectories (nature of response) of GA, PSO, and proposed GWO are slightly

Fig. 11 Current and desired torques at active joint (θ_{11}) with error between them for GA, PSO, GWO algorithms and experimental, respectively

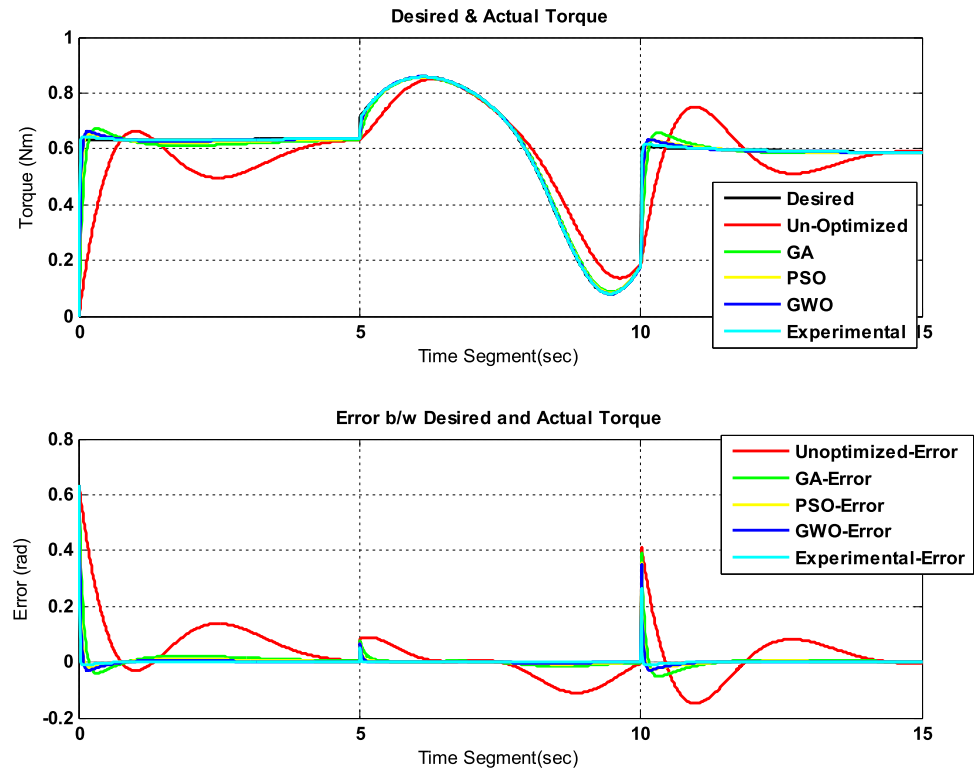


Table 7 Comparison of torque error and fitness value at joint angle (θ_{11})

Torque Performance Indices Comparison				
Error	Legend: ■ Un-Optimized ■ GA ■ PSO ■ GWO ■ Experimental			
	SSE	ISE	IAE	Jmin
■ Un-Optimized	15.62	9.15	11.97	16.77
■ GA	7.22	4.16	5.53	7.74
■ PSO	2.91	1.84	2.06	0.964
■ GWO	0.465	0.005	0.019	0.49
■ Experimental	0.486	0.006	0.021	0.51

Joint Angle (θ_{11})

varied from each other at the same environmental condition. The reason behind is the ability to respond against the disturbances and to optimize the parameters of LQR-PID controller under same conditions.

The simulation results show that the proposed GWO can generate much more optimized controller parameters for generating optimal torque and accurate trajectory tracking control.

7 Conclusion

The optimization of Q and R matrices of LQR regulator and tuning of K_p , K_i , and K_d gain parameters of PID controller is done for optimal generation of torque at the active joints for achieving accurate path tracking control of a Maryland manipulator. To test the optimal path tracking control, three dissimilar and consecutive paths are taken

under observation and control action is performed by LQR-PID controller.

For acquiring optimal matrices, the evolutionary optimization algorithms like GA and PSO are applied, for tuning of PID gain parameter, but both the methods fail to achieve accurate trajectory tracking control as the simulation results show higher values of fitness function (J), sum square error (SSE), integral square error (ISE), and integral absolute error (IAE) between the desired and the actual trajectory at all joint angles, as discussed in the above section.

The proposed gray wolf optimizer (GWO) algorithm is demonstrated by numerical simulation and then experimentally conducted. The results show that the proposed GWO provides better positioning accuracy; the comparison simulation results are highlighted as follows:

- i. Error analysis due to the path tracking control at joint angle (θ_{11}) is such that the SSE in GWO is 63.29% lesser than GA and 42.79% lesser than PSO, the ISE in GWO is 93.75% lesser than GA and 88.70% lesser than PSO, the IAE in GWO is 87.61% lesser than GA and 78.30% lesser than PSO, and the $J_{\min}$ in GWO is 97.71% lesser than GA and 94.84% lesser than PSO.
- ii. Error analysis due to the path tracking control at joint angle (θ_{12}) is such that the SSE in GWO is 67.04% lesser than GA and 37.42% lesser than PSO, the ISE in GWO is 99.46% lesser than GA and 98.80% lesser than PSO, the IAE in GWO is 99.26% lesser than GA and 98.16% lesser than PSO, and the $J_{\min}$ in GWO is 88.0% lesser than GA and 82.63% lesser than PSO.
- iii. Error analysis due to the path tracking control at joint angle (θ_{13}) is such that the SSE in GWO is 58.62% lesser than GA and 11.15% lesser than PSO, the ISE in GWO is 99.83% lesser than GA and 99.59% lesser than PSO, the IAE in GWO is 99.94% lesser than GA and 99.86% lesser than PSO, and the $J_{\min}$ in GWO is 86.96% lesser than GA and 53.77% lesser than PSO.
- iv. Error analysis due to the torque produced at the input joint angle (θ_{11}) is such that the SSE in GWO is 93.55% lesser than GA and 84.02% lesser than PSO, the ISE in GWO is 99.87% lesser than GA and 99.72% lesser than PSO, the IAE in GWO is 99.65% lesser than GA and 99.07% lesser than PSO, and the $J_{\min}$ in GWO is 93.66% lesser than GA and 49.17% lesser than PSO.

Hence, the proposed GWO proves better among all and it provides high accuracy in trajectory tracking control with better performance indices.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

References

- Stewart D (1965) A platform with six degrees of freedom. *Proc Inst Mech Eng* 180(1):371–386
- Li Y, Xu Q (2007) Design and development of a medical parallel robot for cardiopulmonary resuscitation. *IEEE Trans Mechatron* 12(3):265–273
- Shoham M, Burman M, Zehavi E, Joskowicz L, Batkalin E, Kunicher Y (2003) Bone-mounted miniature robot for surgical procedures: concept and clinical applications. *IEEE Trans Robot Autom* 19(5):893–901
- Xu WL, Pap JS, Bronlund J (2008) Design of a biologically inspired parallel robot for foods chewing. *IEEE Trans Industr Electron* 55(2):832–841
- Xu WL, Torrance JD, Chen BQ, Potgieter J, Bronlund JE, Pap JS (2008) Kinematics and experiments of a life-sized masticatory robot for characterizing food texture. *IEEE Trans Industr Electron* 55(5):2121–2132
- Ottaviano E, Ceccarelli M (2006) Application of a 3-DOF parallel manipulator for earthquake simulations. *IEEE Trans Mechatron* 11(2):241–246
- Yoon WK, Suehiro T, Tsumaki Y, Uchiyama M (2004) Stiffness analysis and design of a compact modified delta parallel mechanism. *Robotica* 22(4):463–475
- Ramadan AA, Takubo T, Mae Y, Oohara K, Arai T (2009) Developmental process of a chopstick-like hybrid-structure two-fingered micro manipulator hand for 3-D manipulation of microscopic objects. *IEEE Trans Industr Electron* 56(4):1121–1135
- Berenguer FJ, Huelin FMM (2008) Zappa, A quasi-passive biped walking robot with a tail: Modeling, behavior, and kinematic estimation using accelerators. *IEEE Trans Ind Electron* 55(9):3281–3289
- Pierrot F, Nabat V, Company O, Krut S, Poignet P (2009) Optimal design of a 4-DOF parallel manipulator: from academia to industry. *IEEE Trans Rob* 25(2):213–224
- Huang T, Chetwynd DG, Mei JP, Zhao XM (2006) Tolerance design of a 2-DOF over constrained translational parallel robot. *IEEE Trans Rob* 22(1):167–172
- Ren L, Mills JK, Sun D (2008) Trajectory tracking control for a 3-DOF planar parallel manipulator using the convex synchronized control method. *IEEE Trans Control Syst Technol* 16(4):613–623
- Chaudhary G, Ohri J (2016) “3-DOF Parallel manipulator control using PID controller”, *IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems*, 1–6
- Su YX, Duan BY, Zheng CH (2004) Nonlinear PID control of a six DOF parallel manipulator. *IEEE Proc Control Theory Appl* 151(1):95–102
- Li Y, Wang Y (2009) Trajectory tracking control of a redundantly actuated parallel robot using diagonal recurrent neural network, in *Proceedings International Conference Natural Computing*, Tianjin, China, IEEE, 292–296

16. Hamamci SE (2007) An algorithm for stabilization of fractional-order time delay systems using fractional-order PID controllers. *IEEE Trans Autom Control* 52(10):1964–1969
17. Efe MÖ (2011) Neural network assisted computationally simple $PI^{\lambda}D^{\mu}$ control of a quadrotor UAV. *IEEE Trans Industr Inf* 7(2):354–361
18. Ahmed A (2019) Controllers comparison to balancing and trajectory tracking a two wheeled mobile robot. *Int J Robot Autom (IJRA)* 5(1):28–31
19. Das S, Pan I, Halder K, Das S, Gupta A (2013) LQR based improved discrete PID controller design via optimum selection of weighting matrices using fractional order integral performance index. *Appl Math Model* 37(6):4253–4268
20. Saha S, Das S, Das S, Gupta A (2012) A conformal mapping based fractional order approach for sub-optimal tuning of PID controllers with guaranteed dominant pole placement. *Commun Nonlinear Sci Numer Simul* 17(9):3628–3642
21. Verma SK, Yadav S, Nagar SK (2017) Optimization of fractional order PID controller using grey wolf optimizer. *J Control Autom Elect Syst* 28(3):314–322
22. Razmjoooy N, Ramezani M, Namadchian A (2016) A new LQR optimal control for a single-link flexible joint robot manipulator based on grey wolf optimizer. *Majlesi J Elect Eng* 10(3):53–60
23. Altinoz OT (2018) Multi objective PID controller design for active suspension system: scalarization approach. *Int J Optim Control Theor Appl* 8(2):183–194
24. Oh SK, Jang HJ, Pedrycz W (2011) A comparative experimental study of type-1/type-2 fuzzy cascade controller based on genetic algorithms and particle swarm optimization. *Expert Syst Appl* 38(9):11217–11229
25. Bharadwaj CS, Babu TS, Rajasekar N (2018) Tuning PID controller for inverted pendulum using genetic algorithm. *Adv Syst Control Autom* 442:395–404
26. Wang L, Luo C, Li M, Cai J (2017) Trajectory planning of an autonomous mobile robot by evolving ant colony system. *Int J Robot Autom (IJRA)* 32(4):406–413
27. Sheng L, Li W (2018) Optimization design by genetic algorithm controller for trajectory control of a 3-RRR parallel robot. *Algorithms J* 11(7):2–13
28. Shao X, Zhang J, Zhang X (2019) Takagi-Sugeno fuzzy modeling and PSO-based robust LQR anti-swing control for overhead crane. *Hindawi Math Problems Eng* 2019(21):1–14
29. Şen MA, Kalyoncu M (2019) Grey wolf optimizer based tuning of a hybrid LQR-PID controller for foot trajectory control of a quadruped robot. *Gazi Univ J Sci* 32(2):674–684
30. Tsai LW, Stamper R (1997) A parallel manipulator with only translational degrees of freedom, presented at the Proceedings of ASME, design engineering technology conference, Irvine, CA, USA, 1–18
31. Tsai LW (1999) Robot analysis and design: the mechanics of serial and parallel manipulators. John Wiley, Hoboken, New York, USA, pp 134–142
32. Bolandi H, Esmailzadeh SM (2011) Exact tip trajectory tracking control of a flexible robot arm. *Int J Robot Autom* 26(1):100–109
33. Li S, Wenfeng Wu, Ma D (2018) MPPT of photovoltaic system variable acceleration disturbance method based on genetic algorithm. *Int J Robot Autom (IJRA)* 33(2):179–185
34. Alajlan M, Chaari I, Koubaa A, Bennaceur H, Ammar A, Youssef H (2016) Global robot path planning using GA for large grid maps: modelling, performance and experimentation. *Int J Robot Autom* 31(6):1–12
35. Kennedy J, Sammut C, Webb GI (2011) Particle swarm optimization, encyclopedia of machine learning. Springer, Washington, USA, pp 760–766
36. Panda S, Padhy NP (2008) Comparison of particle swarm optimization and genetic algorithm for FACTS-based controller design. *Int J Appl Soft Comput* 8(4):1418–1427
37. Ekin S (2016) Application and comparative performance analysis of PSO and ABC algorithms for optimal design of multi-machine power system stabilizers. *Gazi Univ J Sci* 29(2):323–334
38. Mirjalili S, Mirjalili SM, Lewis A (2014) Grey wolf optimizer. *Adv Eng Softw* 69(7):46–61
39. Mirjalili S, Saremi S, Mirjalili SM, Coelho LDS (2016) Multi-cost grey wolf optimizer: a novel algorithm for multi-criterion optimization. *Expert Syst Appl* 47(10):106–119
40. Mirjalili S (2015) How effective is the Grey Wolf optimizer in training multi-layer perceptrons. *Appl Intell* 43(1):150–161
41. Shahzad S, Mirjalili SZ, Mirjalili SM (2014) Evolutionary population dynamics and grey wolf optimizer. *Neural Comput Appl* 26(5):1257–1263
42. Sánchez D, Melin P, Castillo O (2017) A grey wolf optimizer for modular granular neural networks for human recognition. *Hindawi Comput Intell Neurosci* 2017(8):1–26
43. Choubey C, Ohri J (2021) Optimal trajectory generation for a 6-DOF parallel manipulator using grey wolf optimization algorithm. *Robotica* 39(3):411–427
44. Ghoreishi SA, Nekoui MA (2012) Optimal weighting matrices design for LQR controller based on genetic algorithm and PSO. *Adv Mater Res* 433–440:7546–7553

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